

Simulation of the behavior of a payload fairing during separation

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Simulation of the behavior of a payload fairing during separation

Degree project in Aerospace Engineering at Advanced Level

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Abstract

The present paper describes the development of a finite element based tool at the launcher directorate of the French Space Agency (CNES) in order to support the ground study and testing of fairing separation systems. The main topic is the finite element modeling process and its validation. The considered finite element methods and resolution algorithms are presented, with a focus on shell element formulations in the used finite element packages. The model results are compared with those obtained for a separation system with a pre-existing model and a ground test. The developed model shows good agreement with the reference results and enables the CNES to fulfill its technical support role.

Keywords: finite-element modeling, modal analysis, non-linear dynamics, transient response

Nomenclature

$\mathbf{C}$	Damping matrix [kg/s]
F	Force [N]
F_1	Initial force [N]
f	Eigenfrequency [Hz]
$\mathbf{G}$	Contact constraints matrix [-]
$G_{zr}, G_{\theta r}$	Transverse shear moduli in cylindrical coordinates [Pa]
g	Gravity [m/s ²]
$\mathbf{K}$	Stiffness matrix [N/m]
k	Spring stiffness [N/m]
$\mathbf{M}$	Mass matrix [kg]
m	Mass [kg]
t	Time [s]
t_b	Locking instant [s]
$\mathbf{u}$	Nodal displacements vector [m]
$\mathbf{x}$	Vector of nodal coordinates for potential contacting nodes [m]
α, β, γ	Damping parameters in the HHT operator [-]
Δl	Elongation [m]
Δu	Vertical elongation [m]
δ_0	Ballistic motion position coefficient [m]
ϵ	Strain [-]
η_0	Ballistic motion speed coefficient [m/s]
σ	Stress [Pa]
Φ	Eigenvector
ω^2	Eigenvalue [(rad/s) ²]

1 Introduction

The finite element (FE) method is a widely used tool in mechanical modeling and nowadays computing power enables research and development centers to tackle large scale complicated problems. Here the problem to be modeled is the separation (also called jettison) in two halves of a payload fairing placed on its ground test facility. The dynamic transient response of the fairing under impulse of the separation system is sought. In Section 2 a global description of the model is given. Section 3 consists of a modal analysis to obtain knowledge of the dynamic properties of the considered fairing. Section 4 covers the functioning mechanical principles of the horizontal separation system model and the response of the fairing to its action is compared to numerical

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reference results and to test results. In Section 5 a similar analysis is conducted for the vertical separation system. Finally Section 6 concludes on the work done and sets recommendations for future studies.

2 Model description

The model represents the fairing on its ground testing support. An overview of it is available in Appendix A. The coordinate systems for the study are presented in Fig. 1 where the $z = 0$ reference for the global Cartesian coordinate system is at ground level. The model is composed of different parts coming from CNES and industrial partners. There exists an industrial model (which will be called the reference model) consisting of the same parts, except for the separation system parts which were modified in accordance with CNES specifications to enable improved simulation of the fairing's behavior during separation.

The separation system, which drives the jettison of the fairing, consists of a horizontal separation system (HSS) and a vertical separation system (VSS). The HSS is located at the bottom of the fairing (see Appendix A) and consists in a two parts ring. The upper ring is attached to the fairing main panel and the lower ring is attached to the cylindrical adapter. The two rings are initially attached together and are separated via a pyrotechnic event. In the FE model the pyrotechnic system is simply modeled with beam elements (between the rings) which are removed to simulate the separation event. In the considered settings for the test to which the FE model is to be compared, a certain number of boxes are attached at the junction of the upper ring and the lower ring. Their role is to lift the fairing before triggering of the vertical separation system.

The size of the model is of about 1.5×10^5 elements and 1×10^6 variables (degrees of freedom and Lagrange multipliers).

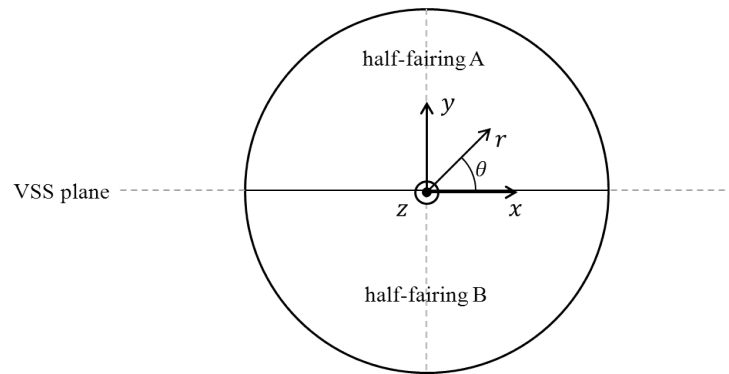


Figure 1: View from the top of the considered coordinate systems.

2.1 Resolution algorithms

The equation to be solved for the general dynamics problem including contact is written as [12] :

$$\mathbf{M}\ddot{\mathbf{u}}_{t+\Delta t} + \mathbf{C}\dot{\mathbf{u}}_{t+\Delta t} + \mathbf{K}\mathbf{u}_{t+\Delta t} + \mathbf{G}_{t+\Delta t}\boldsymbol{\lambda}_t = \mathbf{F}_{t+\Delta t}^{\text{ext}} \quad (1a)$$

$$\mathbf{G}_{t+\Delta t}\mathbf{x}_{t+\Delta t} \leq 0 \quad (1b)$$

which corresponds to a constrained optimization problem. The first line consists in the dynamic equilibrium equation where $\mathbf{u}$ is the vector of nodal displacements (the dot indicates derivation with respect to time), $\mathbf{M}$ is the mass matrix, $\mathbf{C}$ is the damping matrix, $\mathbf{K}$ is the stiffness matrix, $\mathbf{G}$ is the matrix enabling enforcement of the contact constraints, $\boldsymbol{\lambda}$ is the vector of tangential and normal forces and $\mathbf{F}^{\text{ext}}$ is the vector of external forces. The second line is the contact constraint equation where $\mathbf{x}$ is the vector of potentially contacting nodal coordinates. This equation system is solved using the Abaqus [1] FE package.

The FE problem is non-linear in the sense of "boundary" non-linearity [1]. Contact is implemented as well as kinematic couplings such as MPC. There is no material non-linearity. Besides there is no geometric non-linearity. Different tests were conducted in an effort to implement large displacements, but gave no significant results due to issues regarding the contact resolution (see sections 4 and 5).

An implicit scheme, associated with a contact solver are chosen for the time resolution.

Implicit dynamics An implicit method, based on the Newton-Raphson algorithm [10] is used in Abaqus ¹.

¹The reason for not choosing an explicit scheme comes from previous studies which shown a higher sensitivity of explicit schemes with respect to the initial conditions versus implicit schemes.

The operator used to approximate the equation of motion (1a) is the Hilbert, Hugues and Taylor [7] (denoted HHT) algorithm. It enables direct integration of the equation of dynamics and tuning of the numerical damping. It is defined by:

$$\mathbf{M}\ddot{\mathbf{u}}_{t+\Delta t} + (1 + \alpha)\mathbf{K}\mathbf{u}_{t+\Delta t} - \alpha\mathbf{K}\mathbf{u}_t = \mathbf{F}_{t+\Delta t}^{\text{ext}} \quad (2a)$$

$$\mathbf{u}_{t+\Delta t} = \mathbf{u}_t + \Delta t^2 \left[\left(\frac{1}{2} - \beta \right) \ddot{\mathbf{u}}_t + \beta \ddot{\mathbf{u}}_{t+\Delta t} \right] \quad (2b)$$

$$\dot{\mathbf{u}}_{t+\Delta t} = \dot{\mathbf{u}}_t + \Delta t[(1 - \gamma)\ddot{\mathbf{u}}_t + \gamma\ddot{\mathbf{u}}_{t+\Delta t}] \quad (2c)$$

and initial conditions. The three damping parameters α , β and γ can be chosen (see [3]) to specify the characteristics of the scheme. In the present case, the parameters ensures second-order precision and optimal damping of high frequencies modes by having $\alpha \in [-0.5, 0]$ and the other two defined by:

$$\beta = 0.25(1 - \alpha^2) \quad (3a)$$

$$\gamma = 0.5 - \alpha \quad (3b)$$

Two options are retained [1]: $\alpha = -0.414$ by default for medium damping (especially for contact problems) and $\alpha = -0.05$ (for low damping) if explicitly stated.

Contact formulation The contact resolution in Abaqus relies on the following methods:

- Detection of contact: the approach “path-based” algorithm is a finite-sliding one, which enables tracking of potentially contacting nodes throughout the loading history and updating of corresponding mass and stiffness in the assembled matrices. This is feasible even though geometric non-linearity is disabled [1].
- Discretization of contact: “surface-to-surface” discretization (also called “segment-to-segment”). This method has a smoothing, averaging, effect on the contact constraints (in the sense of an optimization problem). It avoids large, undetected, penetrations of master nodes into slave segments.
- Contact constraints enforcement: a penalty-based algorithm is used. It requires to use a penalty coefficient, which corresponds to a contact surface stiffness. In the present case this value is 10 times the equivalent stiffness of the neighboring elements [1].

A pure master-slave frame is used, where the master surface is on the stiffer structure, knowing that both surfaces have similar mesh sizes, and, when applicable, the smaller surface is chosen as the slave. As surface-to-surface discretization is chosen, the choice of master-slave contact pairs have less effect on the results [1], it can however affect the computation time. These aspects of contact modeling can be counted as part of the numerical uncertainty (see section 6).

2.2 On shell elements

A shell is a deformable medium assimilated to a surface with a thickness. It distinguishes from a plate by its intrinsic curvature. Thus one needs to describe the way a surface deforms in space which makes shell elements one of the hardest to formulate. A shell is said to be shallow in a mathematical sense if the curve radius (inverse of the curvature) is large compared to the thickness [5].

Love-Kirchhoff shells Shells in the sense of Love-Kirchhoff are shallow shells using the following hypothesis [5, 6]: the sections, initially orthogonal to the neutral section (or fiber), remain orthogonal after deformation, the strain field is developed to the first order (Taylor) with respect to the transverse parameter. This leads to a plane strain state for the shell and no out-of-plane stress. Elements based on such theory can only describe bending and membrane efforts.

Transverse shear treatment If the strain field is developed to the second order [6] then the sections are allowed not to remain orthogonal and transverse shear effects are taken into account. However it is a crude approximation of the transverse shear field. It can be improved by multiplying the transverse shear modulus by a correction factor. The two classical approaches are Reissner’s [11] and Mindlin’s [9]. Reissner computed the energy developed by the transverse shear in 3D and in plate theory for a plate submitted to pure bending, equated them and obtained a factor of 5/6 (≈ 0.83). Mindlin’s approach [9] is to evaluate the frequency of the first antisymmetric mode of transverse shear vibration in 3D and with a infinite plate, setting them equal to yield a factor of $\pi^2/12$ (≈ 0.82).

Shells in Abaqus and Nastran The four nodes quadrilateral elements S4/S4R in Abaqus and CQUAD4 in Nastran rely on a similar formulation (see [4, 8]). The element is isoparametric, assumes displacement shapes for in-plane behavior, and assumes linear transverse shear strain to overcome shear-locking. They have four integration points and can use reduced integration (one point at the center of the element) as is the case for S4R, which requires stabilization of spurious modes [1]. The transverse shear computation relies on Reissner–Mindlin type approach and should be used with great care when studying transverse shear behavior (see Appendix B for an example).

3 Modal analysis

3.1 The eigenvalue problem

In order to validate the global dynamic properties of the payload fairing model, a free-free modal analysis is conducted and the results are compared to the known modal characteristics of the half-fairings. The equation to be solved is [1]:

$$[\omega^2 \mathbf{M} + \omega \mathbf{C} + \mathbf{K}] \Phi = 0 \quad (4)$$

where ω^2 is the eigenvalue and Φ is the associated eigenvector, which is mass-normalized when extracted. For easier interpretation, the eigenfrequency $f = \omega/2\pi$ will be used. The 14 first eigenfrequencies are extracted from both half-fairings and the complete fairing using the Lanczos eigensolver from Abaqus [1]. For this analysis all parts that usually interact through contact are merged by having coincident nodes.

3.2 Verification of the modeling

Since it is a free-free analysis 6 rigid-body modes are expected with eigenfrequencies close to zero before the flexible modes. With the use of "mesh-independent fasteners" [1] to link the HSS upper ring to the fairing main panel in version 6.10 of Abaqus, the 6 rigid modes were not obtained and large variations were observed in the eigenfrequencies when tuning these fasteners. The exact same input data used with Abaqus version 6.11 gave a more physical modal basis with 6 rigid modes. Hence, unless stated otherwise, all computations are made with version 6.11 of Abaqus.

Besides, the HSS upper ring was modeled in solid elements (it was in shells and beams in the reference model) and it was necessary to cancel the transverse shear moduli $G_{\theta r}$ and G_{zr} in the material formulation, otherwise the flexible modes 1 and 2 of the half-fairings would be in inversed order with wrong eigenfrequencies (with respect to the reference model). In this case, the thin-shell behavior is the one documented in the reference model. This was confirmed by a VSS impulse response computation (see section 5). One hypothesis to explain this phenomenon is a compatibility issue between the shell elements nodes degrees of freedom (5 dof) and those of the solid elements (3 dof).

3.3 Study of the modal basis

The modal behavior of the fairing halves without boxes being known (from previous experiences and proven models), the modal basis obtained with the HSS upper ring modeled with solid elements should be close. Prior knowledge of this kind of systems indicates that a variation of 5% of the eigenfrequencies, with respect to the reference model, is acceptable as maximum tolerance.

In Table 1, the 4 first eigenfrequencies of the half-fairing A are compared to those of the reference model. The conclusion is that the values are within the limitations and this validates the modal basis. In Table 2 one can see the variation of frequencies due the presence of the boxes (the modes being the same as previously). When looking at the first mode, it is interesting to note that despite their small size compared to the diameter of the fairing, the boxes do not act as point masses. From the spring-mass formula $\sqrt{k/m}$, a point-mass effect would lower the frequency, but here the frequencies increase. This stiffening effect (increase of the frequency) is confirmed when observing the density of elastic strain energy in Fig. 2. The location of the maximum density corresponds to the location of some of the boxes. Such analysis was also conducted on other modes of the half-fairing and of the other half-fairing and gave similar conclusions.

Flexible mode number	Reference model normalized frequency [Hz]	Current model variation [%]
1	1	-2,80
2	1	0,47
3	1	-0,08
4	1	-3,84

Table 1: Comparison of eigenfrequencies of half-fairing A from reference model and current model.

Flexible mode number	Current model normalized frequency (no boxes) [Hz]	Frequency variation due to boxes [%]
1	1	2,51
2	1	-0,51
3	1	-0,09
4	1	1,58

Table 2: Impact of the boxes on the modal basis of half-fairing A.

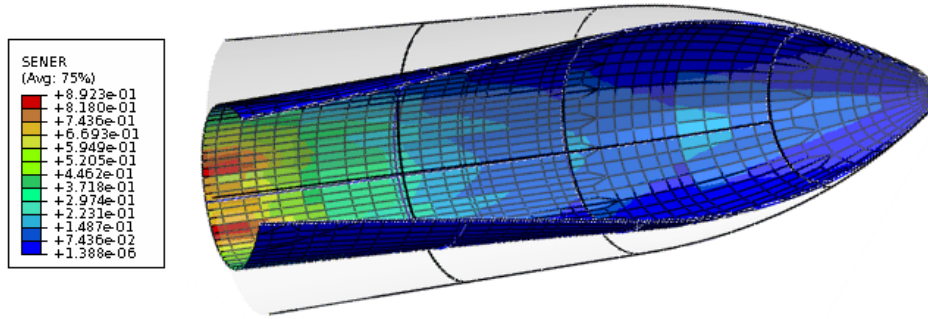


Figure 2: Elastic strain energy density on mode 1 of half-fairing A.

4 Analysis of the behavior during horizontal separation

4.1 Description of the horizontal jettison

The kinematic behavior of the FE modeling of these boxes is shown in Fig. 3. The improvement brought in regards of the reference model is that the guiding device and the locking device were modeled in details. In the reference model these systems were simulated through kinematics conditions and rigid elements. In the present model, their geometric features are accurately represented with 3D solid elements and the necessary contact conditions are implemented. Contact is also implemented between the foot and the lower-support.

The idea in the model is to have an elastic connector element which stores elastic deformation energy. As the beam elements at the support junctions are removed, the connector elements elastically unloads and, as the lower support is clamped, lifts the upper part. The locking device plays a crucial role in the considered model. As it stops the extension of the connector, it transmits, via reactions forces, a part of the inertia accumulated in the upper part. Hence, at the interface between the foot and the lower support, the reaction of the lower-support towards the foot is greater than the weight of the upper part, which results in the lifting of this upper part off the lower support.

4.2 Finite element procedure

As explained in the earlier section, the connector element is to act elastically with pre-compression and its line of action should follow the displacement of the connector's ends during deformation. There are two ways to model this in Abaqus.

The first idea uses an "axial connector" [1] with an elastic behavior and a "constitutive reference length" to ensure pre-compression. If this length is not specified, the free length is the geometric distance between the two points defining the connector. However this method implied divergence of the computation due to contact resolution.

The second idea is a type A Spring element (SpringA) [1] with a non-linear elastic behavior defined by $F = -F_1 + k\Delta l$ in order to have a non-zero force when the elongation is zero. This solution was proven compatible with contact. However, it requires to simulate the effect of the pre-compression (initial deformation) on the supporting structures with a couple of opposed forces before including the SpringA.

The whole assembly is loaded through different steps:

- steps 1 and 2: the different parts (all but the connector elements SpringA) of the model are assembled (static step) and the gravity is applied linearly as a ramp (static step);
- step 3: couples of opposed forces are applied linearly as ramps to points A and B (SpringA end points) until the desired level is obtained (static step). This creates the initial deformation of the structure.;
- step 4: the beams at the HSS rings junction are removed instantly. The model stabilizes numerically in a time very shot compared to the next step. It avoids having to balance this perturbation in parallel of the contacts and displacements of the next step (dynamic step);

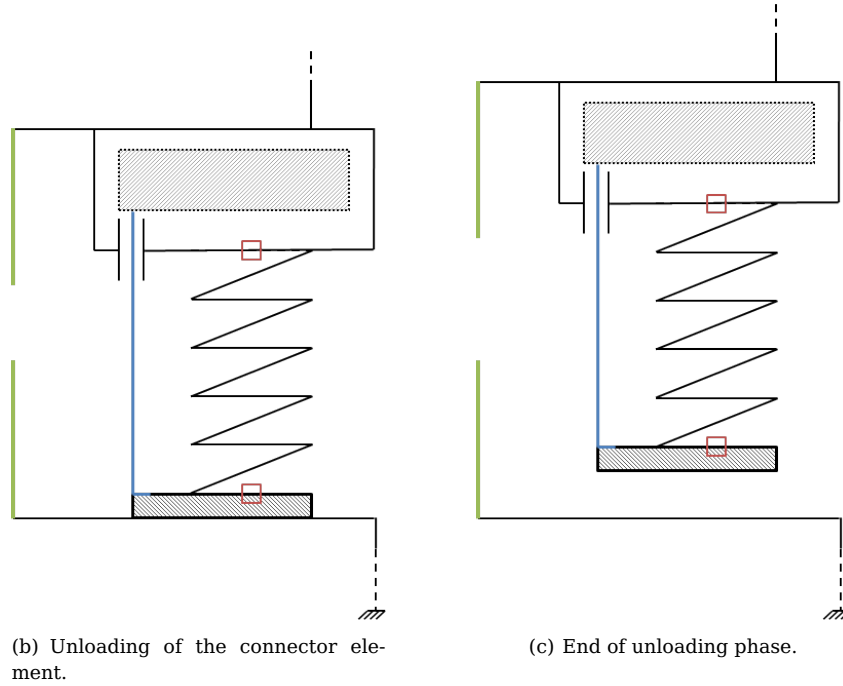
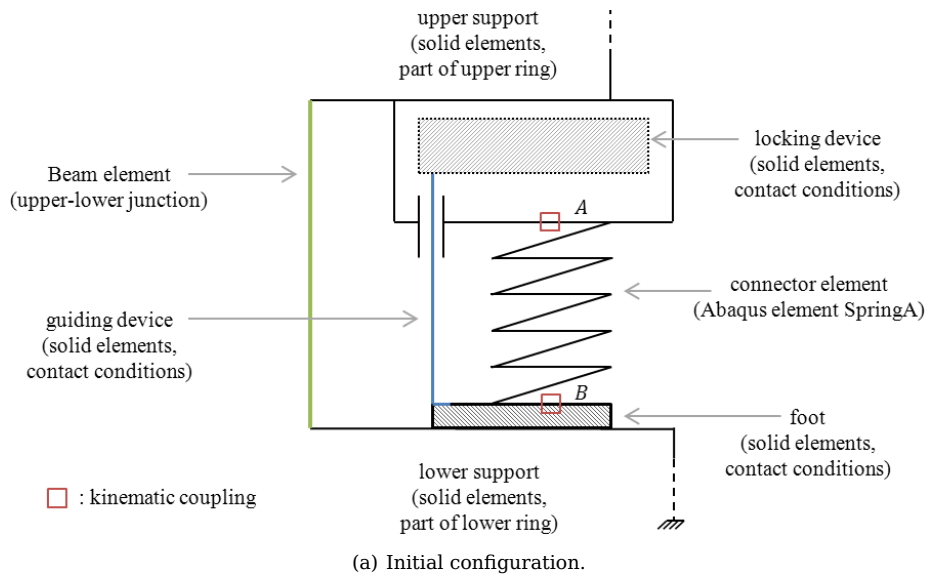


Figure 3: Scheme of the finite element model describing the kinematics. In (a) the lower and upper support are attached. In (b) the beams are removed and the compressed Abaqus element SpringA element unloads. In (c) the locking device has stopped the unloading and the accumulated inertia lifts the upper support.

- step 5: the couples of opposed forces are removed instantly and the SpringA are inserted in the model. They unload until locking and the ballistic phase is obtained (dynamic step).

In all steps, the feet of the support tower are clamped.

To enable comparison and to validate the FE model, an analytical study is conducted.

4.3 Analytical study

The fairing is considered as a rigid body here and the modeled system can thus be thought of as a spring-mass system for the first phase and as a ballistic shot for the second phase. For the first phase the equation of motion reads as

$$m\Delta\ddot{u} = mg + F_1 - k\Delta u \quad (5)$$

where Δu is the vertical (along the axial direction of the fairing) elongation of the connector element, F_1 is the initial force applied due to pre-compression of all the connector elements, k is the equivalent spring stiffness and g is gravity. Solving using initial conditions gives

$$\Delta u(t) = \frac{mg - F_1}{k} \cos\left(t\sqrt{\frac{k}{m}}\right) + \frac{F_1 - mg}{k} \quad (6)$$

At time t_b when the locking device activates, the equation of motion becomes

$$\Delta\ddot{u} = -g \quad (7)$$

The solution is then

$$\Delta u(t) = -\frac{1}{2}gt^2 + \eta_0 t + \delta_0 \quad (8)$$

where η_0 and δ_0 are determined respectively from the speed and the position at t_b given by (6).

4.4 Results and discussion

In Fig. 4 the vertical displacement of the fairing, taken at the location of one of the test sensors (close to one of the boxes), is compared to the analytical results and to the test results. The time and the displacement are normalized with respect to their test values. The comparison between the analytical and FE results indicates that the assumption of rigid body behavior for the fairing and the supports is not valid.

Verification The concept of verification is defined by [2]: “The process of determining that a model implementation accurately represents the developer’s conceptual description of the model and the solution to the model.”

The FE model is verified as it agrees with the theory (analytical modeling equations (6) and (8)). Indeed it simulates qualitatively the same lifting behavior as the analytical model. The mesh, the geometrical and material characteristics were checked *a priori*. After computation, the correct behavior of the SpringA (stress-strain relation) was obtained which confirms the input data. Besides, the contact force at the foot of the lower support is shown in Fig. 5, normalized with respect to the expected value. It tells us that due to the choice of a SpringA combined with a pre-loading step and a short intermediary one to remove the junction beams, there is 3% “overshoot” in the initial SpringA push value.

Validation The concept of validation is defined by [2]: “The process of determining the degree to which a model is an accurate representation of the real world from the perspective of the intended uses of the model.”

Validation then comes from the comparison of the FE results to the test results. At normalized time $t = 1$ there is a 4% difference between the FE computations and the test results in Fig. 4. To obtain an overall accuracy assessment, such differences are computed for all boxes and the average is obtained: 11%. According to CNES specifications, these figures are good enough and form a first step towards complete validation of the FE model. On the other hand, the analytical model is not validated as it has a 20% difference at normalized time $t = 1$ with respect to the test results taken at the top of the fairing.

To enable further validation, the uncertainty on the test measurements should be quantified, especially for the average error assessment. In addition, an increased understanding of the physical behavior of the model can serve a better interpretation.

Energy balance It is interesting to understand the distribution of energy stored in the SpringA elements towards the fairing and to the adjacent structures (lower ring, cylinders, support tower and platform) during the step 5, and this taking into account the work of the weight. In the analytical analysis, all the elastic potential energy of the SpringA is transferred as kinetic energy to the fairing, while in the FE model, it is also transferred as strain energy.

In order to obtain this distribution, the following method was used: at each instant t , the amounts of strain energy and kinetic energy in the fairing and the adjacent structures are computed, the sum of the works produced by the SpringA and the weight are evaluated, and the maximum later value is used to normalize the energies of the fairing and the structures. This gives an approximate image of the distribution of the energy delivered by the SpringA. The result is shown in Fig. 6. It is an approximation because the energies of the fairing and adjacent structures include the effect of the weight (this effect being available as a whole model effect only, through the work it produced during step 2). One can say from this graph that in the beginning of the SpringA push, the adjacent structures take most of the energy, in strain energy form; then the transfer gives more energy to the fairing, in kinetic form. The moment when this latter energy is maximum corresponds to the triggering of the locking device (the SpringA no longer drives the fairing’s motion). From this point, the fairing is only subject to its weight, which works against the motion, thus its kinetic energy decreases.

In an effort to check the global energy balance of the system, the energy delivered by the SpringA and the work of the weight is plotted along with the mechanical energy of the whole structure excluding the SpringA. This is shown in Fig. 7. The analysis is as follows: the system $\mathcal{S} = \{\text{fairing and adjacent structures}\}$ is submitted to the sum of the works of the SpringA action and the weight. Hence the variation of the mechanical energy of $\mathcal{S}$ should theoretically be equal to the work of the external forces (SpringA and weight actions). Though we can see they are fairly close, there is a 3% gap at normalized time 0.51, which is the instant at which the SpringA

have delivered all the energy they could deliver. This value can serve as an uncertainty measure of the FE computation in terms of global energy estimation. Thus one can say that, in this FE computation, $89\% \pm 3\%$ of the external work is transmitted to the fairing, while $8\% \pm 0.3\%$ is transmitted to the adjacent structures. This analysis is valid only if the energy dissipation (including both mechanical and numerical dissipations) is small in comparison to the aforementioned energy discrepancy, which is true for the present study. Besides, during the whole procedure, the artificial energy (from hourglass stabilization in reduced integration shell elements) used to stabilize the computation is kept at a level low enough to be disregarded.

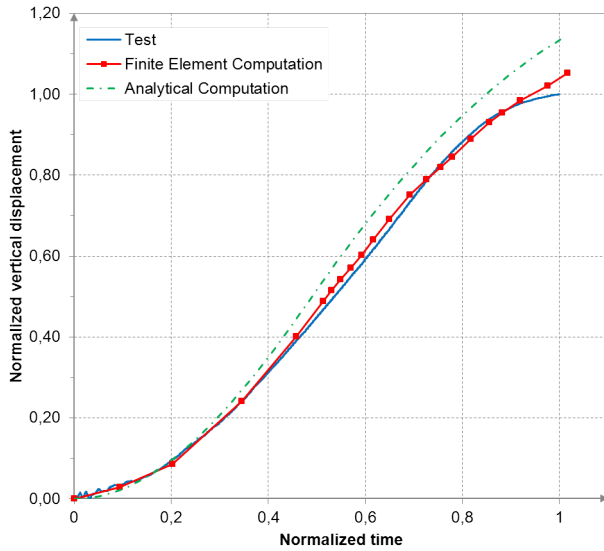


Figure 4: Comparison of vertical displacements.

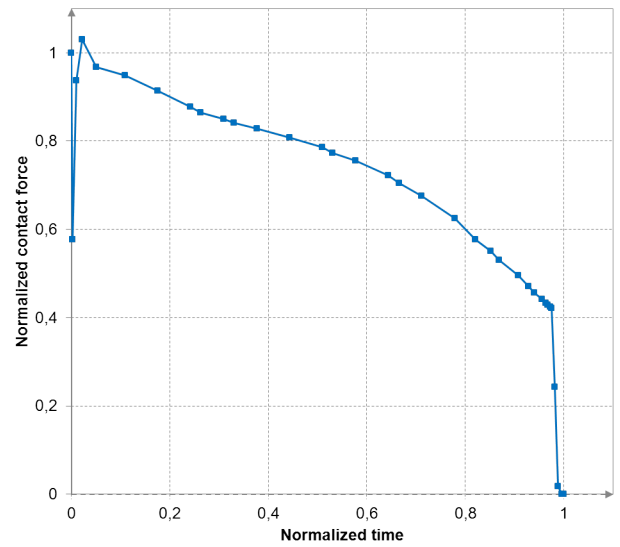


Figure 5: Contact force from $t = 0$ to lift-off time.

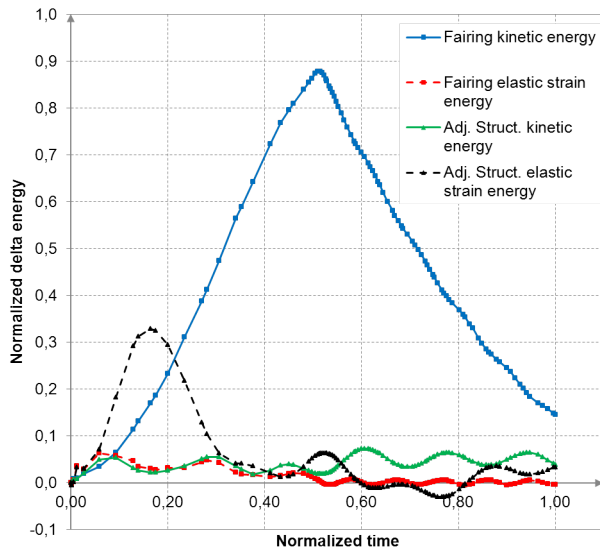


Figure 6: Distribution of energy in the fairing and the adjacent structures.

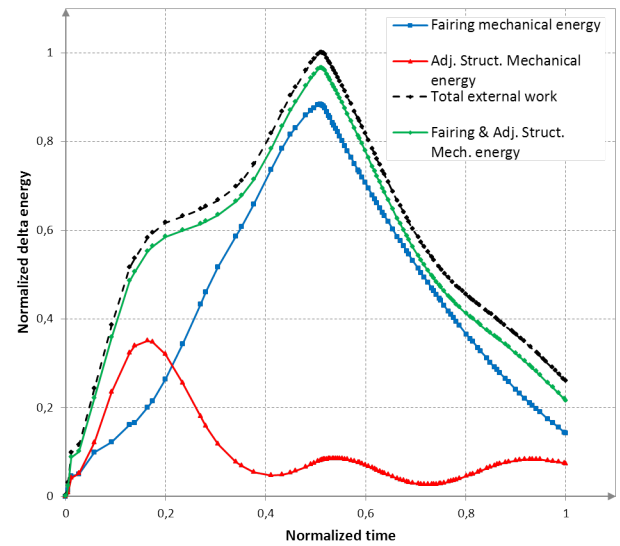


Figure 7: Energy balance relative to the maximum external work.

5 Analysis of the behavior during vertical separation

At a certain time t_1 after having triggered the HSS system, the VSS is triggered. In this part the responses of both half-fairings to the VSS impulse are analyzed in two cases: when the fairing is at rest (no previous action other than gravity) and right after the end of the HSS phase.

5.1 Description of the vertical jettison

The vertical jettison corresponds to a pyrotechnic event taking place in the plane xz which separates the fairing into two halves. It pushes the half-fairings A and B respectively in the $y \geq 0$ direction and the $y \leq 0$ direction. In the reference model, the junction between the two fairings was modeled with beams elements and the impulse

was defined via non-linear type 2 spring connectors elements (the line of action does not rotate with the displacement of the element defining points). Their behavior was set to reproduce the energetic content per unit length of the pyrotechnic system.

In the considered model, the Spring2 elements were replaced with elastic axial connectors so that the line of action accounts for the displacements on the connector's end points. Their behavior was unchanged, as it already complied with CNES specifications.

Under action of the VSS, the first flexible mode of the fairing is the main excited mode. This gives a time scale for the desired duration of the procedure as one full cycle is sought for the analysis.

5.2 FE procedure

According to the case to be studied, one or two steps are applied:

- step 1 (only applied if preceded by HSS phase): the junction beams between the two half-fairings are removed. As for the HSS study, this is a numerical artifice to enable better stabilization of the computation (dynamic step) ;
- step 2: the axial connectors are inserted in the model and the system is let free to evolve during 1.15 times the first flexible mode's period (dynamic step).

As mentioned in Section 4, all contacts in the HSS locking and guiding devices were implemented, which requires heavy computation to solve. In order to avoid excessive computation times and/or output database sizes that would lead to abortion, it was necessary to remove the guiding devices from the boxes. It was first attempted to alter or remove the contact pairs, though it was not allowed by Abaqus. It was thus thought that removing the devices was easier and would not affect much the responses. This is discussed in the next subsection.

5.3 Results and comments

The projection on the x-axis of the displacement of one corner of a half-fairing is shown in Fig. 8. It corresponds to an image of the first flexible mode excited during this procedure.

Verification The behavior of the connectors was checked for correct interpretation of the input. The shape of the curves in Fig. 8 corresponds to the excitement of the flexible mode 1, which is expected from the industrial reference. The response of the half-fairings to the VSS action (without previous HSS action) was used to confirm that the alteration of the HSS upper-ring material was necessary. The mode excited under action of the VSS if the upper ring has an isotropic material is the right one in terms of shape but the still has a wrong period (from the wrong eigenfrequency, see section 3).

The validation comes from the comparison of the computation results with the test results, which were not available to the author at the time of the writing. Besides, given the large displacements (in terms of rigid body motions for instance) observed during this phase, it is the opinion of the author that geometric non-linearity should be activated for this phase. This was not achieved due to convergence issues. One hypothesis is local buckling-like behavior of the shell elements near the VSS action as may indicate the large difference between the average force seen in the model and the maximum residual force value ². Despite of using Euler scheme based algorithms [1] designed to overcome such issues, convergence was not obtained by the author.

Influences of the guiding device and the HSS phase From Fig. 8 there is a 0.8% difference between the VSS responses with and without guiding devices. The curves including guiding devices can thus be used for direct comparison with curves excluding guiding devices. There is a 1.5% difference between the response to the VSS action preceded by the HSS action and the response to the sole VSS action. Hence one can see that the HSS action has little influence on the response to the VSS action. Due to the absence of geometric non-linearity, the VSS phase preceded by the HSS phase should be interpreted as the effect of superposed HSS and VSS actions on the considered displacement (except for contacting parts which are treated in finite-sliding).

Influence of the numerical parameters Two parameters are studied here: the damping intensity of the HHT scheme and its automatic time increment variation. Figure 9 shows the responses with a medium damping ($\alpha = -0.414$) and a small damping ($\alpha = -0.05$), and this with a fixed rule for the variation of the time increment. One can see that there is no significant influence of this parameter in the present case (1.2% difference). On the other hand, one can see that the rule for automatic variation of the time increment causes a difference in the displacement, which is of 7.8%³ on the average on the whole procedure. This figures are given for the mentioned computation conditions and should be revised in accordance with the improvement of these conditions.

²This idea came from discussion of the author with the Abaqus support team

³See section C in the appendix for a discussion on this figure.

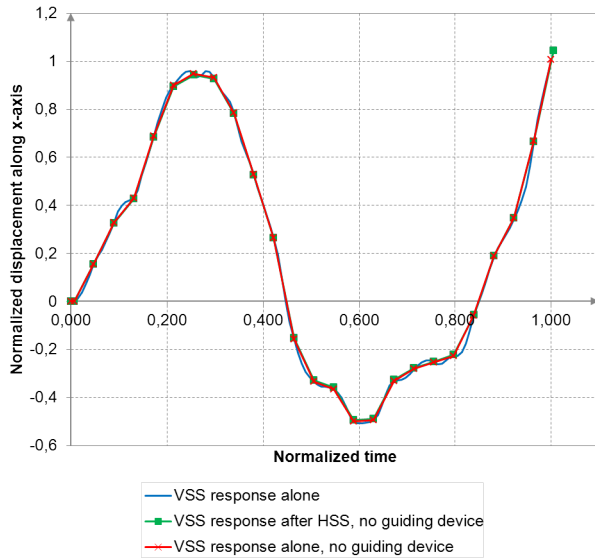


Figure 8: Displacement of one corner of half-fairing A – influence of the guiding device.

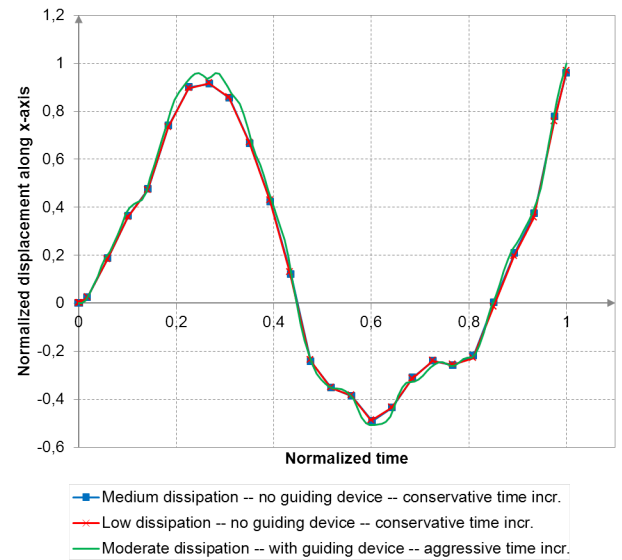


Figure 9: Displacement of one corner of half-fairing A – influence of time step incrementation and numerical damping.

6 Conclusion and recommendations

6.1 Conclusion

The aim of the paper was to present the development of a FE based tool for the simulation of ground fairing separation tests. After having presented the theory used in the FE software and the different assumptions it implied, the fairing as studied.

The model is a large model in terms of number of degrees of freedom as it implies contact resolution. This contact resolution enabled improvement of the modeling of the boxes for the HSS action simulation. On the other hand it renders large displacements (geometric non-linearity) hard to take into account.

The modal analysis required modifications to match the reference industrial document. Given the results obtained for the HSS action modeling it seems acceptable to say that small displacements may be kept if one is interested only in the HSS response since the amplitude is small compared to the typical length of the model. In terms of comparison with the test, the values obtained before fine tuning of the model and include of the measurement uncertainty are satisfactory.

The VSS action modeling required removing the guiding devices in order to overcome contact related issues. A first analysis was conducted and showed the influence of the integration algorithm parameters. To enable further investigation, an effort will be necessary in order to implement geometric non-linearity.

6.2 Recommendations

For future studies on the subject the author's recommendations are:

- for shell elements, given that the Abaqus code is a proprietary code, access to all the features of the formulations is not an easy task. Though there is a big experience in the use of Nastran in the space field, it is not the case for Abaqus (and yet due its capacities, it is usually used if a highly non-linear computation involving dynamical events, contacts and/or multi-physics is needed, indicating a need for higher precision and reliability). Hence a benchmark of the code on selected internal (industrial) cases with known, proven, results could improve knowledge in this field.
- as for the modeling of the HSS action and the corresponding test results:
 - it is the author's opinion that a reliable uncertainty measurement for the test results would serve a more accurate interpretation;
 - from the point of view of the modeler, two choices can apply: either the geometry is refined along with the mesh, which could enable better treatment of the contacts and avoid singularities occurring when activating geometric non-linearity, or the model is simplified. Three phenomena are accurately modeled: the guiding (orientation of the action), the stopping device and the contact (force transfer modeling). It could be interesting to isolate each of these phenomena in an effort to see which one has most influence on the improvement of the results with respect to the industrial reference model. In both cases, the mesh of the fairing main panel in the vicinity of the boxes should be refined for better

representation of local effects (if this is done, a re-meshing of the HSS upper could help minimizing the use of mesh-independent fasteners);

- for better matching of the test, a sensitivity analysis should be conducted on: the SpringA elements (stroke length, force inputs), pre-constraint state of the fairing used for the test (stiffness dispersion of composite parts), and friction coefficients between contacting parts of the boxes;
- for numerical uncertainty assessment, the effects of the following should be investigated: the geometric non-linearity, the contact modeling (though the purpose is not accurate resolution of the local contact rather than its impact on the overall behavior), the integration algorithm parameters of damping, residual tolerances and time increment variation (especially if large displacements is achieved), a more accurate representation of the boundary conditions at the bottom of the tower, a more accurate representation of the catching devices of the fairing (modeled as point masses as of now, stiffness could be taken into account), the correlation between the position of the measuring devices and the nodes at which the FE output is obtained ⁴.

- for the modeling of the VSS action:

- accounting for large displacements should be a priority.
- the modeling of the VSS action through nonlinear spring-like elements may be a contributing factor. A simulation with a more physical acceleration-like input could help determining this point;
- due to the high number of beam and shell jointures in the model, mesh compatibility and potentially unstable degrees of freedom should be studied. This would require a pre-processor more powerful (given the size and complexity of the model) than the one available to the author.

Acknowledgments

The author would like to thank first the French Space Agency (CNES) for its financial and technical support and for having welcomed him in the Structures, Thermics and Materials department of the Launcher Directorate. The author would like to thank especially RUAG Space for having supplied some meshes and the test results and ALTEN for its technical support in meshing of the model and in establishing the procedure of the first calculations. Finally the author would like to express his gratitude towards his advisors, Gunnar Tibert at KTH and Benoit Tang at CNES, for their guidances, their supports and the knowledge he acquired thanks to them.

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⁴In the present case, this was identified as a second order effect.

A Model overview

The Fig. 10 represents the assembly of the finite element model and shows the main parts constituting the model.

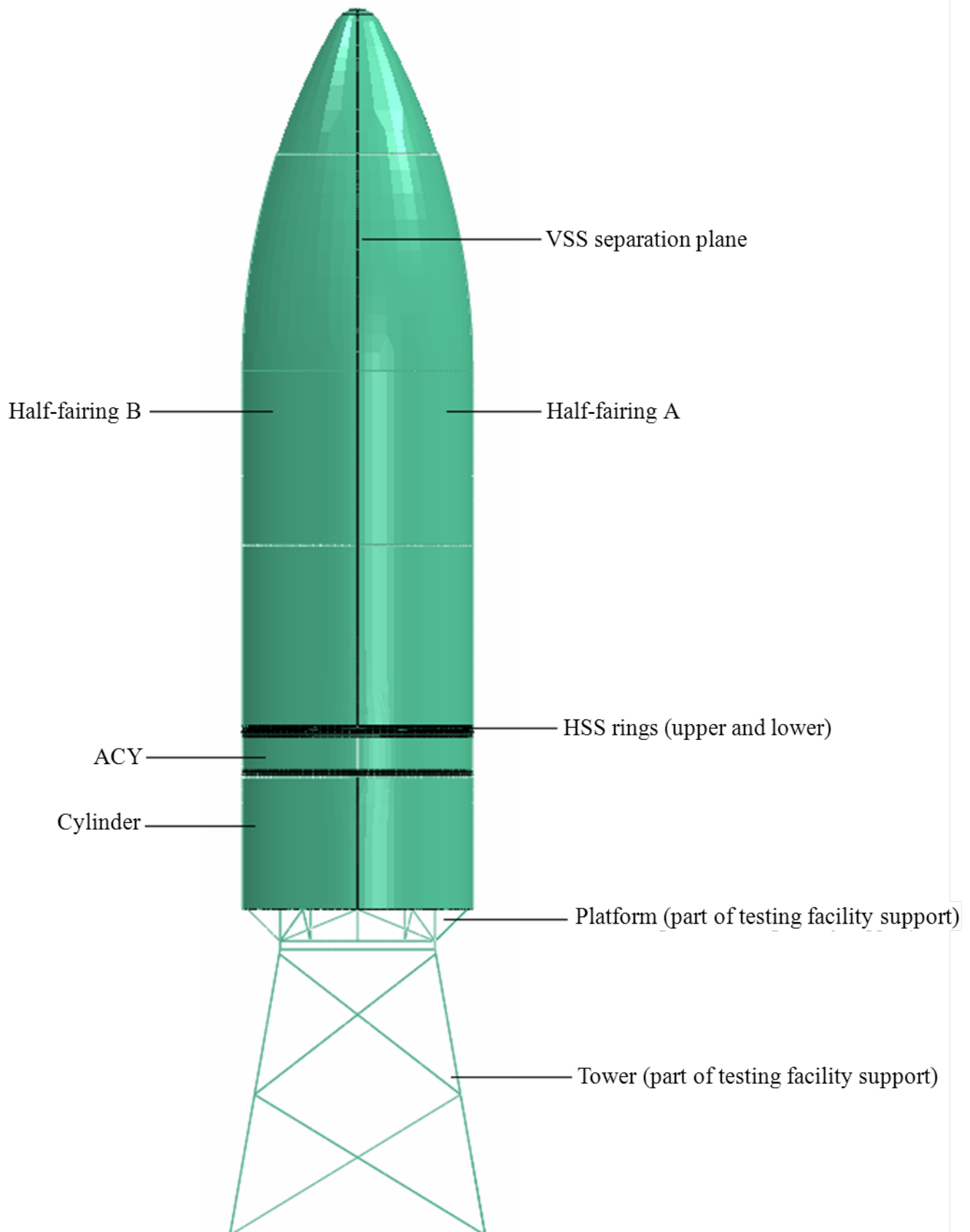


Figure 10: View of the whole finite element assembly for the studied model.

B Example: A shell submitted to transverse shear

The example proposed here corresponds to a shell submitted to a load case such that transverse shear phenomena can not be neglected. It aims at showing the limitations of Nastran and Abaqus classical shell elements⁵ in regards of transverse shear.

The shell and its loading can be seen in Fig. 11. It has the following characteristics (only the relative values matter here which is why no unit is needed in this case): the curve radii in directions 1 and 2 are respectively 25.0 and infinity, the thickness is 0.25, the Young's modulus is 4.32×10^8 and the Poisson's ratio is 0. One straight edge of the shell is clamped, and its other straight edge is constrained to displace only in the vertical plane (in particular, rotations degrees of freedom are constrained). The load is a force in the vertical direction. The resolution is linear and is conducted with Nastran (due to the high similarity with Abaqus' elements in this case). Three models are proposed:

- a Love–Kirchhoff model which corresponds to CQUAD4 elements with a very low (10^{-6}) correction factor. This gives a 2D model not made to account for transverse shear;
- a Reissner–Mindlin model which corresponds to CQUAD4 elements with the default value (5/6) for the correction factor. This corresponds to 2D a model accounting for transverse shear (via rotating but straight sections);
- a solid elements model with four elements in the transverse direction. This gives a model that can allow for non-straight sections. It is used as a reference for the present case.

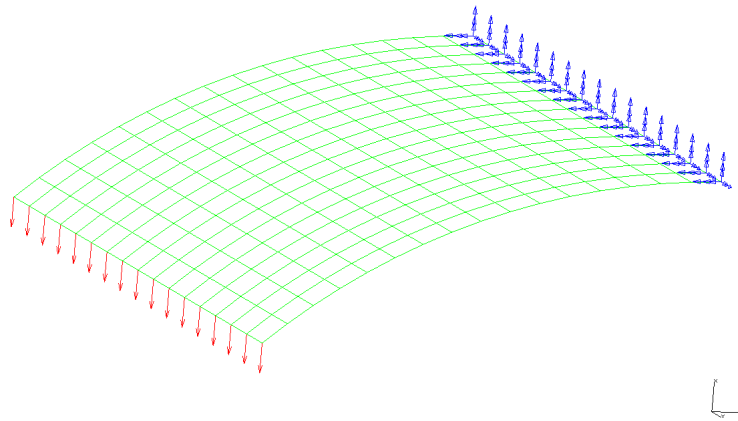


Figure 11: Shell elements transverse shear test case.

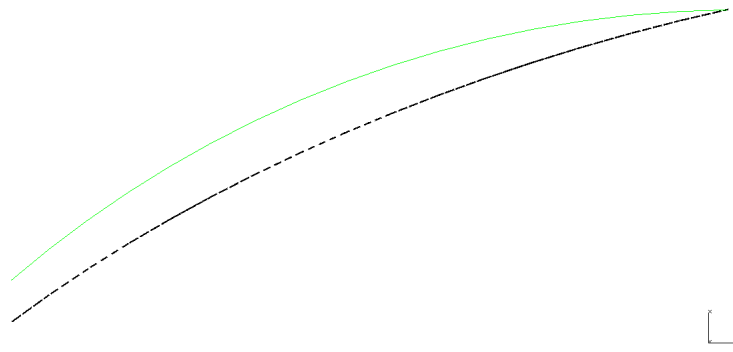
The maximal deflections are normalized with respect to the solid model result, which is thus equal to 1.0, in order to enable comparison. The Reissner–Mindlin model has a maximal deflection of 0.81 and the Love–Kirchhoff model gives a value of 1.2×10^3 . The deformed shapes corresponding to these deflections are shown in Fig. 12.

As expected, the Love–Kirchhoff shell gives unacceptable results. The Reissner–Mindlin shell has a 20% difference with the solid results. A sensitivity study was conducted on the correction factor and it was found that no significant change could be observed for values greater than 0.5, as it can be seen in Table 3.

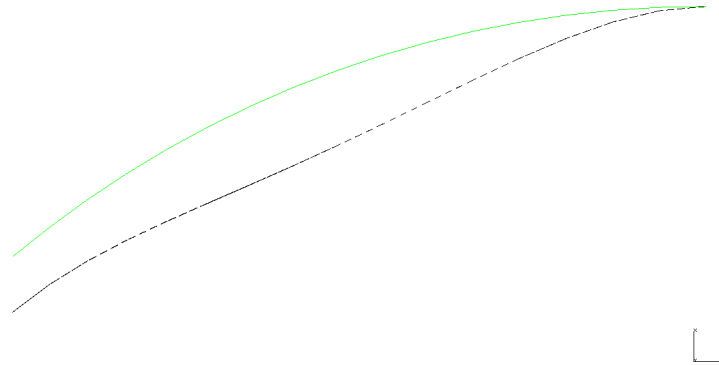
This example shows that the way transverse shear is approximated is not always appropriate and that there may be no suitable value for the correction factor.

One conclusion is that such transverse shear treatment should be used as an improvement for the overall behavior of a shell structure but not for precise transverse shear evaluation. This is especially true if thick shell structures such as sandwiches are considered.

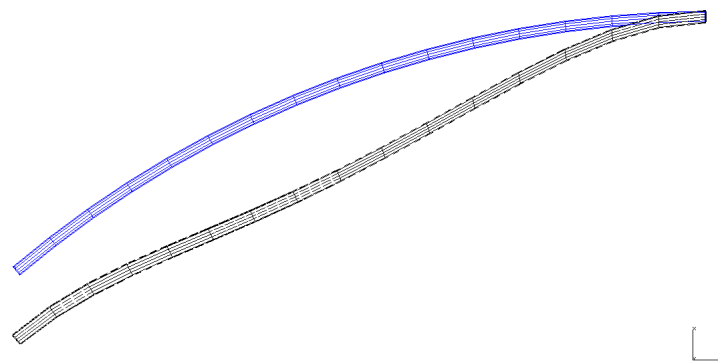
⁵For more information regarding the evolution of shell elements in classical codes, the interested reader may refer to "Perspective on finite elements for shell analysis" by R. H. MacNeal in *Finite Elements in Analysis and Design*, vol. 30, pages 175–186, 1998.



(a) Shallow shell, Love-Kirchhoff type.



(b) Thick shell, Reissner-Mindlin type.



(c) Solid elements shell with 4 elements through the thickness.

Figure 12: Comparison of deformed shapes.

Correction factor	Normalized maximal deflection
0.	1479.42
0.0001	16.26
0.001	2.65
0.01	1.17
0.1	1.01
0.5	1.00
0.833 (=5/6)	1.00
1.	1.00

Table 3: Influence of the correction factor on the maximum deflection (normalization with reference to the solution for a 5/6 correction factor).

C Discussion on the average relative error of the VSS phase FE results

When studying the influence of the numerical parameters on the displacement curve for the VSS phase (see section 5), the author first used the post-treatment module Abaqus/Viewer [1]. This module interpolates between values when two curves with different output frequencies are compared. In our case, the curved used as a reference (VSS action after HSS action) has a five times larger output frequency than the others. The reasons why the output frequency was lowered are the time of the computation and the size of the output databases.

The way Abaqus/Viewer interpolates is:

- it first takes the union of the abscissas points of both considered curves to create a global output abscissa set ;
- then for each of the points in this global set, it interpolates, for both curves, a value (ordinate) point if there is not one already.

In our case this causes two issues. The first one is that though the the output frequency is changed, the computation precisions (time increments) are still close, so it can artificially degrade the curve with the lower output frequency. The second one is that the observed displacements cancel three times, thus causing very large relative errors close to these cancellations. The relative error computed this way is shown in Fig. 13.

Remark *This precision regarding the post-treatment is given here because the rule for the variation of the time increment was modified, which gave different output instants, necessitating to interpolate the results that are not given at the desired instants.*

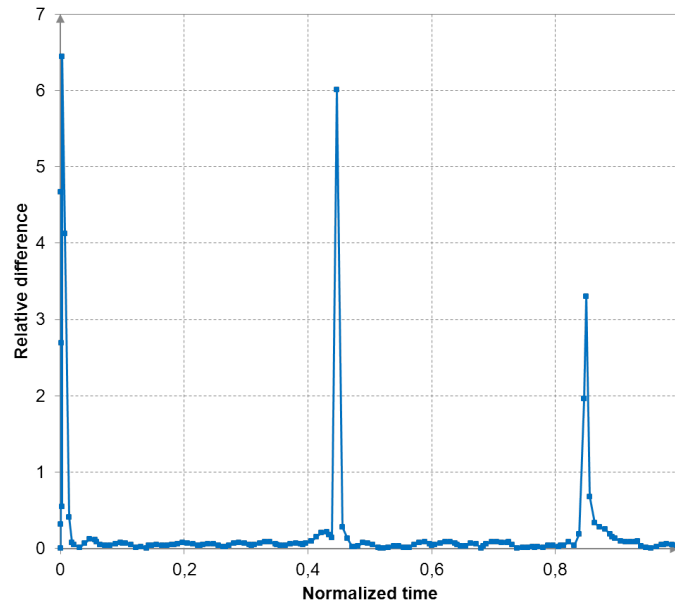


Figure 13: Relative error computed with Abaqus/Viewer.

In an effort to minimize the choice of which points should be disregarded in the computation, another post-treatment was applied. The set of abscissas which corresponds to the smallest output frequency curve is used to linearly interpolates values on the curve with the largest output frequency. The relative error is then computed for this reduced number of points. The operation is shown in Fig. 14. The resulting relative error is shown in Fig. 15. One can see that this method avoids numerical singularities due to zero crossings. If the point at normalized time 0.85 is excluded from the average computation, then the figure is 7.8%. If the points at normalized times 0.002 and 0.43 are also excluded (they also correspond to zero crossings regions) then the average is 5.5%. To be conservative, the largest value is kept in the main text.

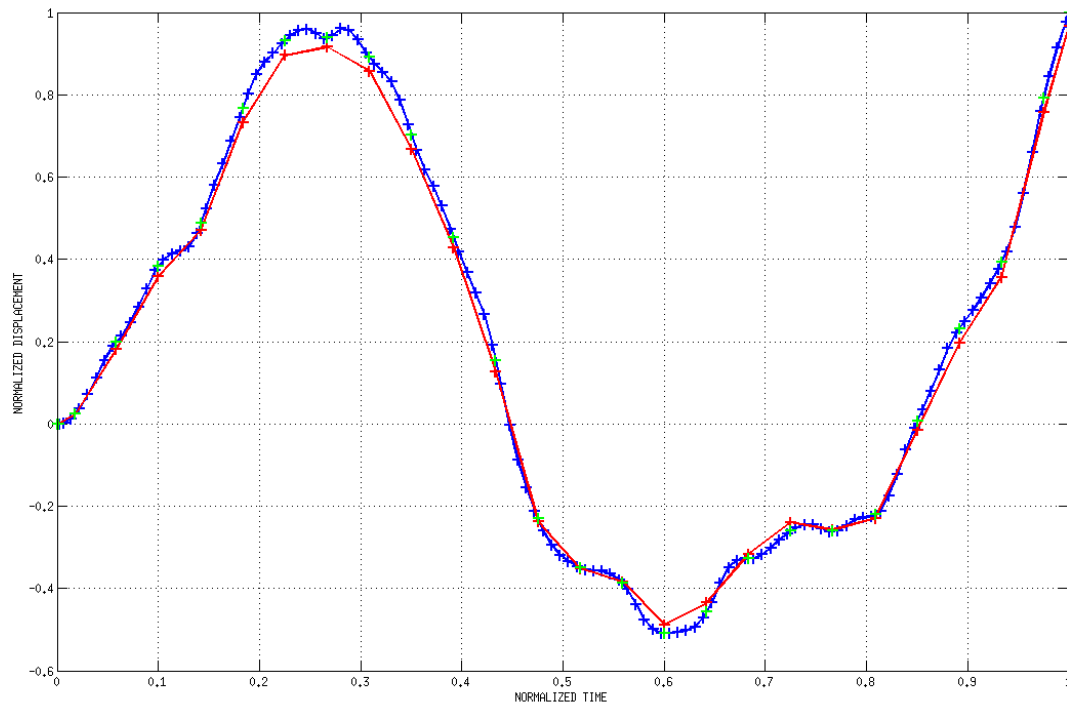


Figure 14: Interpolation operation. The blue curve corresponds to the reference curve, *i.e.* the VSS action after the HSS action. The red curve is the VSS action response obtained with the conservative time increment variation rule and the green + markers are the points interpolated to enable comparison.

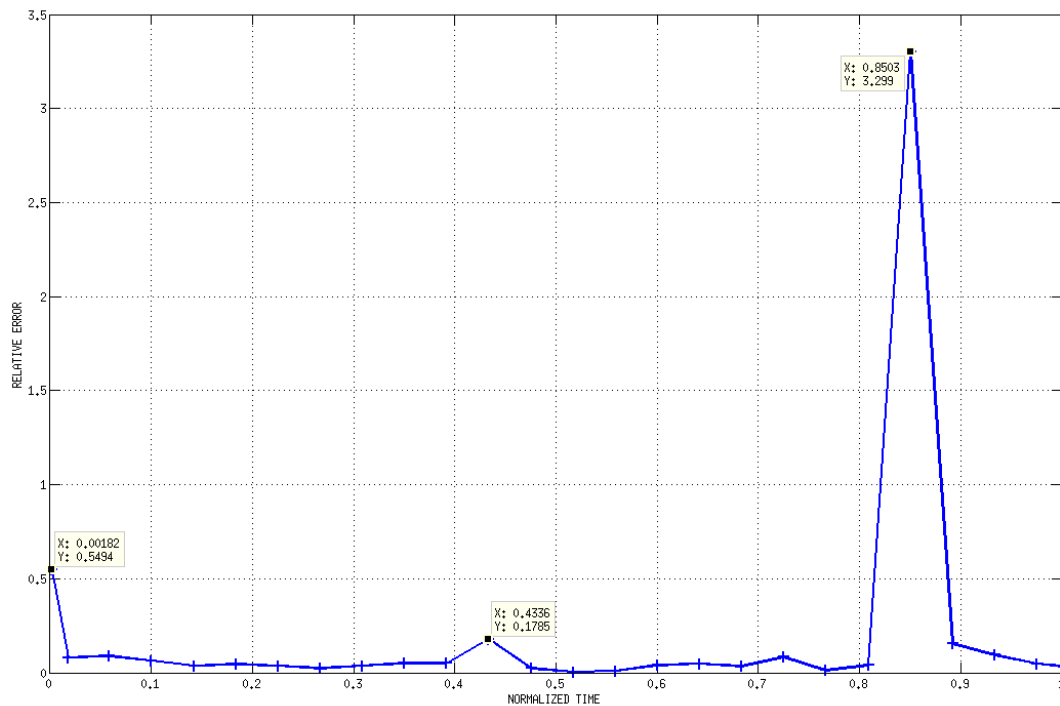


Figure 15: Relative error computed with conservative linear interpolation.